

each to 1in (25cm) long, per areole. From spring to early summer, bears funnel-shaped, bright red flowers, 3in (7cm) across, with prominent, red stamens and green stigmas. Z11–12. (min. 45°F/7°C) **var. paucispinus** illus. p.496.

Echinofossulocactus. See *Stenocactus*.
Echinomastus macdowellii. See *Thelocactus macdowellii*.

ECHINOPS

Globe thistle

ASTERACEAE/COMPOSITAE

Genus of annuals, biennials and perennials, grown for their globelike, spiky flower heads. Does best in full sun and in poor soil. Propagate by division or seed in fall, or by root cuttings in winter. Flower heads dry well.

E. bannaticus. Upright, clump-forming perennial. H 4–5ft (1.2–1.5m), S 2ft (60cm). Has narrow, deeply cut leaves and pale to mid-blue heads of spherical flowers, borne on branching stems in late summer.Z5–9. ‘**Taplow Blue**’ illus p. 241.

E. ritro ‘Veitch’s Blue’. Upright perennial. H to 3ft (90cm), S 18in (45cm). Sharply divided, dark green leaves are downy beneath. Round, thistlelike, purplish-blue heads of flowers are borne on silvery stems in late summer. Z3–9
E. sphaerocephalus. Massive, bushy perennial. H 6ft (2m), S 3ft (1m). Has deeply cut, mid-green leaves, pale gray beneath. Gray stems bear round, grayish-white flower heads in mid- to late summer. Z3–9

ECHINOPSIS

SYN. LOBIVIA, TRICHOCEREUS

CACTACEAE

Genus of spherical to columnar, perennial cacti, mostly freely branching; sometimes held to include *Trichocereus*. Requires full sun and well-drained soil. Propagate by seed or offsets in spring or summer.

E. aurea, syn. *Lobivia aurea*, *L. cylindrica*, *Pseudolobivia aurea*, illus p.495.

E. backebergii, syn. *Lobivia backebergii*, illus. p.491.

E. candicans, syn. *Trichocereus candicans*, illus. p.492.

E. chamaecereus, syn. *Cereus silvestrii*, *Chamaecereus silvestrii*, *Lobivia silvestrii* (Peanut cactus), illus. p.486.

E. cinnabarina, syn. *Lobivia cinnabarina*. Spherical, perennial cactus. H and S 6in (15cm). Glossy, dark green stem has about 20 warty ribs and mostly curved, dark spines. In summer bears funnel-shaped to flattish, carmine-red flowers, 3in (8cm) across. Z13–15

E. eyriesii. See *E. oxygona*.

E. lageniformis, syn. *Trichocereus bridgesii*, illus. p.489.

E. marsoneri, syn. *Lobivia haageana*, illus. p.491.

E. multiplex. See *E. oxygona*.

E. oxygona, syn. *E. eyriesii*, *E. multiplex*, *Echinocactus eyriesii*, illus. p.481.

E. pentlandii, syn. *Lobivia pentlandii*, illus. p.491.

E. rhodotricha. Spherical to columnar, perennial cactus. H 2ft (60cm), S 8in (20cm). Produces branching, dark green stems,

3½in (9cm) across, with 8–13 ribs. Curved, dark spines, ¾in (2cm) long, later turn pale. Has tubular, white to pink flowers in spring–summer. Z13–15

E. spachiana, syn. *Cereus spachianus*, *Trichocereus spachianus* (Torch cactus), illus. p.492.

Echioides longiflorum. See *Arnebia pulchra*.

ECHIUM

BORAGINACEAE

Genus of annuals and evergreen shrubs, biennials and perennials, grown for their flowers, borne mainly in summer. Requires full sun and fertile, well-drained soil. Water containerized specimens freely in summer, moderately at other times. Propagate by seed in spring or by greenwood or semiripe cuttings in summer. Whitefly may sometimes be troublesome. ☉ All parts may cause mild stomach upset if ingested; contact with the foliage may irritate skin.

E. bourgaeaenum. See *E. wildpretii*.

E. candicans, syn. *E. fastuosum* (Pride of Madeira). Open, usually rounded, woody-based biennial. H 5–8ft (1.5–2.5m), S 5–6ft (1.5–2m). Has rosettes of lance-shaped, white-downy leaves, 6–10in (15–25cm) long. Narrowly funnel-shaped, white, blue-white, or deep purple-blue flowers, are borne in dense flower spikes, to 1ft (30cm) long, from spring to summer. Z10–11 (min. 41–45°F/5–7°C)

E. fastuosum see *E. candicans*.

E. pininana, syn. *E. pinnifolium*. Biennial or short-lived perennial. H to 12ft (4m), S 3ft (90cm). Rosettes have lance-shaped, silver-hairy leaves, to 3in (7cm) long. Each rosette bears a 5–12ft (1.5–4m) long spike of funnel-shaped, blue flowers with large bracts in mid- and late summer. Z8–11.

‘**Snow Tower**’ has white flowers.

E. pinnifolium. See *E. pininana*.

E. vulgare (Viper’s bugloss). Moderately fast-growing, erect, bushy annual or biennial. H 1ft (30cm), S 8in (20cm). Has lance-shaped, dark green leaves. Spikes of tubular flowers, in white, pink, blue or purple, are produced in summer. Z3–8
E. wildpretii, syn. *E. bourgaeaenum*. Evergreen, woody-stemmed, unbranched, erect, biennial or short-lived perennial that dies after fruiting. H 8ft (2.5m) or more, S 2ft (60cm). Narrowly lance-shaped, silver-haired leaves, 1ft (30cm) long, form a dense rosette. Has compact spires, 3–5ft (1–1.5m) long, of small, funnel-shaped, red flowers in late spring and early summer. Z9–10

EDGEWORTHIA

Paper bush

THYMELAEACEAE

Genus of deciduous or evergreen shrubs, grown for their flowers in late winter and early spring. Flowers are susceptible to frost. Best grown against a south- or west-facing wall in most areas. Requires full sun and well-drained soil. Dislikes being transplanted. Take semiripe cuttings in summer or propagate by seed in fall.

E. chrysantha, syn. *E. papyrifera*, illus. p.126.

E. papyrifera. See *E. chrysantha*.

EDRAIANTHUS

Grassy bells

CAMPANULACEAE

Genus of short-lived perennials, some evergreen; sometimes included in *Wahlenbergia*. In winter, a small bud is just visible from each rootstock. In spring, prostrate stems radiate to bear leaves and flowers. Suitable for rock gardens, screes and troughs. Needs sun and well-drained soil. Propagate by softwood cuttings from side shoots in early summer or by seed in fall.

E. dalmaticus. Upright, then arching perennial. H 4in (10cm), S 6in (15cm). Bears narrowly lance-shaped, pale green leaves and, in early summer, terminal clusters of bell-shaped, violet-blue flowers, 1in (25cm) across. Z8–9

E. pumilio, syn. *Wahlenbergia pumilo*, illus. p.368.

E. serpyllifolius, syn. *Wahlenbergia serpyllifolia*, illus. p.367. ‘**Major**’ is an evergreen, prostrate perennial. H ½in (1cm), S to 2in (5cm). Produces tight mats of tiny, ovate, dark green leaves and bell-shaped, deep violet flowers, ¾in (1.5cm) across, on very short stems, in early summer. Needs a sheltered site. Seldom sets seed. Z4–9

EHRETIA

BORAGINACEAE

Genus of deciduous and evergreen shrubs and trees, grown for their foliage and star-shaped flowers. Susceptible to frost damage when young. Requires sun or semishade and fertile, well-drained soil. Propagate by softwood cuttings in summer.

E. dicksonii illus. p.88.

ELAEAGNUS

ELAEAGNACEAE

Genus of deciduous or evergreen shrubs and trees, grown for their foliage and small, usually fragrant flowers, and ornamental fruits. Evergreens provide good shelter belts or hedging, particularly in coastal areas. Most evergreen species thrive in sun or shade, but silver-leaved and deciduous species prefer full sun. Requires fertile, well-drained soil. Trim hedges in late summer. Propagate by seed in fall, evergreen forms also by semiripe cuttings in summer, deciduous forms by softwood or semiripe cuttings in summer.

E. angustifolia (Oleaster) illus. p.116.

E. x ebbingei. See *E. x submacrophylla*.

E. macrophylla. Evergreen, bushy, dense shrub. H and S 10ft (3m). Broadly ovate leaves are silvery-gray when young, becoming glossy and dark green above, but remaining silvery-gray beneath, when mature. Fragrant, bell-shaped, creamy-yellow flowers, silvery outside, are borne from mid- to late fall, followed by egg-shaped, red fruits. Z7–9

E. pungens ‘Maculata’, syn. *E. pungens* ‘Aureovariegata’, illus. p.119.

E. x submacrophylla, syn. *E. x ebbingei*. Evergreen, bushy, dense shrub. H and S 15ft (5m). Has oblong to ovate, glossy, dark green leaves, silvery beneath. Bears fragrant, bell-shaped, silvery-white

flowers from mid- to late fall. Z6–10. ‘**Gilt Edge**’ has eaves with golden-yellow margins. ‘**Limelight**’ illus. p.139.

E. umbellata illus. p.113.

ELAEOCARPUS

ELAEOCARPACEAE

Genus of evergreen, spring- and summer-flowering shrubs and trees, grown for their flowers and foliage. Requires full sun or partial shade and fertile, well-drained but not dry soil. Water containerized plants freely when in growth, less in winter. Current season’s growth may be cut back in winter. Propagate by seed in spring or by semiripe cuttings in summer. Red spider mite and whitefly may cause problems.

E. cyaneus, syn. *E. reticulatus* (Blueberry ash). Evergreen, rounded shrub or tree. H and S 10ft (3m), sometimes to 40ft (12m) or more. Frost tender. Bears elliptic to lance-shaped, toothed, lustrous leaves and, in summer, bell-shaped, fringed, white flowers, and globular, blue fruits in fall. Z14–15 (min. 41°F/5°C)

E. reticulatus. See *E. cyaneus*.

ELEOCHARIS

CYPERACEAE

See also GRASSES, BAMBOOS, RUSHES, and SEDGES.

E. acicularis (Needle spike-rush).

Evergreen, rhizomatous, perennial sedge. H to 4in (10cm), S indefinite. Basal, mid-green leaves are very narrow. Hairless, unbranched, square stems bear solitary tiny brown spikelets in summer. Z9–11

ELEUTHEROCOCCUS

SYN. ACANTHOPANAX

ARALIACEAE

Genus of deciduous shrubs, trees, and sometimes climbers, grown for their foliage and fruits. Produces tiny, usually gree nish-white flowers. Prefers full sun and needs well-drained soil. Propagate by seed in spring or by root cuttings in late winter.

E. sieboldianus, syn. *Acanthopanax sieboldianus*, illus. p.138.

Elliottia paniculata. See *Tripetaleia paniculata*.

ELSHOLTZIA

LAMIACEAE/LABIATAE

Genus of annuals, perennials and semi-evergreen or deciduous shrubs and subshrubs, grown for their flowers. Needs full sun and fertile, well-drained soil. Cut back old shoots hard in early spring. Propagate by softwood cuttings in summer.

E. stauntonii. Deciduous, open subshrub. H and S 5ft (1.5m). Sharply toothed, mint-scented, dark green leaves turn red in fall. Slender spires of pale purple blooms are produced in late summer–fall. Z5–8

Elymus arenarius. See *Leymus arenarius*.